



Scala Programming Practical

Practical – 26

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30) Cluster a dataset into two groups using k-means clustering in Breeze.

Code:

build.sbt

libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"

import breeze.linalg._

object Demo extends App {

// Dataset

val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)

// Initial cluster centers

var center1 = 2.0

var center2 = 10.0

// Repeat clustering

for (i <- 1 to 5) {

**val cluster1 = data.toArray.filter(x =>
math.abs(x - center1) < math.abs(x - center2)
)**

**val cluster2 = data.toArray.filter(x =>
math.abs(x - center2) <= math.abs(x - center1)
)**

// Calculate new centers

center1 = cluster1.sum / cluster1.length

center2 = cluster2.sum / cluster2.length

```
}  
// Display clusters  
println("Cluster 1:")  
println(data.toArray.filter(x =>  
    math.abs(x - center1) < math.abs(x - center2)  
).mkString(", "))  
  
println("Cluster 2:")  
println(data.toArray.filter(x =>  
    math.abs(x - center2) <= math.abs(x - center1)  
).mkString(", "))  
  
println("Center 1 = " + center1)  
println("Center 2 = " + center2)  
}
```

Output:

```
Cluster 1:  
1.0, 2.0, 3.0  
Cluster 2:  
10.0, 11.0, 12.0  
Center 1 = 2.0  
Center 2 = 11.0  
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```